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Model Predictive Control for chiller plant energy efficiency: Leveraging multi-horizon forecasting and machine learning-based system models

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ABSTRACT

Model Predictive Control (MPC) for chiller plants promises significant energy savings but faces practical barriers in retrofit applications, particularly the disconnect between equipment output measurements and actual building thermal state. This paper presents a novel approach using Chilled Water Return (CHR) temperature predictions to enable proactive chiller control. The key innovation lies in developing polynomial CHR prediction models that explicitly incorporate future cooling capacity as inputs, allowing the MPC to evaluate thermal consequences of capacity reductions before traditional thresholds are reached. Unlike reactive control that waits for temperature indicators, our approach predicts CHR evolution under different shutdown scenarios, identifying safe windows for early capacity reduction while ensuring thermal comfort. The framework employs a multi-horizon ensemble with 24 models for load forecasting and 12 models for CHR prediction, providing robust predictions despite discontinuous training data common in real building operations. The resulting Mixed-Integer Nonlinear Programming (MINLP) problem, incorporating Gordon-Ng chiller models and operational constraints, is solved in real-time using Basic Open-source Nonlinear Mixed INteger programming (BONMIN) with automatic fallback strategies. Field deployment in a 5000 RT commercial building in Bangkok demonstrates 11.07% energy savings compared to conventional control. Extended validation over 43 days confirms consistent performance with weather-independent savings. The approach leverages existing temperature sensors already present in every chiller plant, making advanced optimization immediately accessible for retrofit applications without additional instrumentation.

1. Introduction

Chiller plants are among the most energy-intensive systems in large commercial buildings, consuming more than 40% of total energy for space conditioning [1]. In tropical climates, buildings use about 56% of their energy for HVAC [2], with some regions reaching 70% [3]. With global energy consumption projected to increase nearly 50% by 2050 [4] and buildings accounting for 30% of global final energy consumption [5], optimizing chiller plant operations represents a critical opportunity for carbon reduction and cost savings.

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Nomenclature

CHR	Chilled Water Return Temperature
CHS	Chilled Water Supply Temperature
MPC	Model Predictive Control
MINLP	Mixed-Integer Nonlinear Programming
RT	Refrigeration Tons
HVAC	Heating, Ventilation, and Air Conditioning
BMS	Building Management System
IPMVP	International Performance Measurement and Verification Protocol
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error

The complexity of chiller plant optimization arises from nonlinear equipment performance, time-varying loads, interconnected subsystems, and operational constraints. Traditional rule-based control strategies, while robust, result in suboptimal energy consumption due to their reactive nature and inability to anticipate future conditions.

Model Predictive Control (MPC) has emerged as a powerful optimization tool due to its ability to forecast future system states and proactively manage control actions under constraints [6–15]. Over the past decade, numerous studies have demonstrated MPC's potential to reduce energy consumption through optimal chiller sequencing and setpoint control, with reported savings ranging from 5%–14% in simulation environments.

However, practical MPC implementation faces a fundamental challenge: the disconnect between chiller tonnage readings and actual building thermal state. Tonnage measurements reflect instantaneous equipment output influenced by pump dynamics and flow variations, occurring 10–20 min before thermal impacts reach occupied spaces. Building thermal mass further complicates control by causing tonnage oscillations while space conditions remain stable. Most critically, tonnage provides no predictive information about future thermal evolution.

Chilled Water Return (CHR) temperature offers a direct solution to this measurement challenge. CHR reflects the building's aggregate thermal state—water returning from all cooling coils carries the actual heat rejected from occupied spaces. Unlike tonnage, CHR exhibits smooth, predictable dynamics driven by building thermal mass, making it ideal for anticipatory control. Since CHR sensors exist in every chiller plant for safety monitoring, no additional infrastructure is required. By developing predictive models that incorporate future cooling capacity as inputs, we transform CHR from a reactive safety threshold into a proactive control opportunity, enabling early shutdown decisions while ensuring comfort.

This paper develops a predictive control approach using CHR temperature to enable proactive chiller decisions. While conventional control reacts to CHR thresholds, our approach predicts future CHR evolution under different control scenarios, identifying opportunities to reduce capacity early while ensuring comfort. The key insight is CHR's thermal inertia: after reducing cooling capacity, CHR rises gradually, providing a window for optimization.

Our contributions include:

1. A polynomial CHR prediction framework that models the relationship between cooling capacity decisions and resulting temperature trajectories, enabling safe proactive shutdowns.
2. A multi-horizon ensemble approach with 24 load forecasting models and 12 CHR prediction models for robust anticipatory control.
3. Field validation in a 5000 RT chiller plant demonstrating consistent energy savings through real-time MINLP optimization using Basic Open-source Nonlinear Mixed INteger programming (BONMIN).

Unlike approaches requiring additional instrumentation or complex system models, this framework leverages existing sensors to make advanced optimization immediately accessible for the vast inventory of existing buildings requiring energy efficiency retrofits. By transforming CHR from a safety constraint into a predictive control asset, our approach enables sophisticated optimization without the traditional barriers to MPC adoption.

The paper is organized as follows: Section 2 reviews related work and identifies research gaps. Section 3 presents the methodology including load forecasting, CHR prediction, and MPC formulation. Section 4 describes the building system and control architecture. Section 5 analyzes field deployment results including forecasting performance and energy savings validation. Section 6 discusses conclusions and implementation insights.

2. Related work

The application of advanced control strategies to chiller plants has evolved significantly over the past decade. We organize our review into three categories: (1) MPC formulations and optimization approaches, (2) forecasting methods for building energy systems, and (3) real-world implementations and their limitations.

Table 1
Comparison of key MPC studies for chiller plant control.

Study	Method	Forecast type	Validation	Savings	CHR-based
Coffey (2010)	MILP	Load	Simulation	7%–10%	No
Huang (2015)	MINLP	Load	Simulation	5.6%	No
Fan (2022)	Multi-freq	Weather	Simulation	14.3%	No
Kim (2022)	MILP	Load+Solar	Real building	10%	No
This work	**MINLP**	**Load+CHR**	**Real building**	**11.07%**	**Yes**

2.1. MPC formulations for chiller plants

MPC has become prominent for enhancing energy efficiency in chiller plants. Early work by Coffey [6] proposed simulation-based MPC for campus-scale plants, demonstrating 7%–10% energy savings through coordinated control between chillers and thermal storage, though implementation assumed perfect state information.

Ma et al. [10] developed centralized and distributed MPC strategies for building cooling systems. Their centralized approach showed superior performance but required complete system information, while distributed strategies offered better scalability. In subsequent work [15], they extended the framework to include stochastic MPC for handling prediction uncertainties.

Huang et al. [7] developed an MPC-based optimal chiller sequencing strategy, demonstrating 5.6% annual energy savings in simulation. Their MINLP formulation captured discrete ON/OFF decisions and nonlinear performance curves but required up to 10 min solve time—impractical for real-time implementation with 15-minute control intervals.

Recent studies have refined MPC strategies using various optimization methods. Deng et al. [13] combined dynamic programming with MILP to reduce computational complexity while maintaining near-optimal solutions. Their hierarchical approach separated long-term scheduling from short-term control, achieving reasonable solve times but still requiring extensive system modeling.

Borja et al. [14] introduced an economic MPC framework using quadratic programming for parallel chiller operations, reporting energy savings up to 5.19%. By incorporating time-of-use electricity pricing, they demonstrated cost optimization beyond simple energy minimization. However, their study was limited to controlled simulation environments.

Advanced control strategies have also explored machine learning integration. Lee et al. [11] incorporated ANN-based forecasting with occupancy and dynamic pricing signals into their MPC framework. Wang et al. [12] developed a nonlinear MPC strategy for coupled chiller-AHU systems, addressing interdependencies between air-side and water-side systems.

Fan et al. [8] implemented a multi-frequency MPC strategy decomposing control decisions into different time scales, reporting 14.3% hourly energy reduction in simulation—among the highest savings in the literature. However, their approach required perfect coordination between multiple control layers.

2.2. Forecasting methods and real-world implementations

Accurate prediction of cooling loads and system states is fundamental to MPC performance. While the MPC literature typically employs regression models, autoregressive models, or artificial neural networks, these approaches generally focus on single-model predictions without considering forecast uncertainty or horizon-specific characteristics.

Kim et al. [9] represents one of the few field validations, employing MILP-based MPC in a large chiller plant integrated with thermal storage and photovoltaic systems, achieving 10% peak demand reduction. They utilized weather forecasts and historical patterns for load prediction.

The use of extensive ensemble forecasting — such as our approach employing 24 parallel models for hourly load prediction and 12 models for CHR forecasting — is notably absent in existing literature. Single-model approaches dominate despite known limitations in handling measurement noise, non-stationary building usage patterns, and varying forecast accuracy across different time horizons.

2.3. Research gaps and our contributions

Table 1 summarizes key MPC studies for chiller plant control and highlights the unique aspects of our approach.

Our analysis reveals three critical gaps:

Gap 1: Measurement-control disconnect. Previous studies assume cooling load measurements directly represent building demand. However, chiller output responds to water loop dynamics rather than actual thermal requirements, creating a 10–20 min lag between capacity changes and space impacts. This temporal disconnect prevents accurate predictive control.

Gap 2: Reactive control paradigm. Existing strategies, including MPC implementations, remain fundamentally reactive—adjusting capacity based on current conditions or near-term forecasts. No prior work has developed models that predict thermal consequences of capacity reductions, preventing proactive optimization during part-load conditions.

Gap 3: Limited forecast robustness. Single-model predictions fail to capture uncertainty inherent in building operations. Without horizon-specific optimization and uncertainty quantification, MPC implementations risk constraint violations or excessive conservatism.

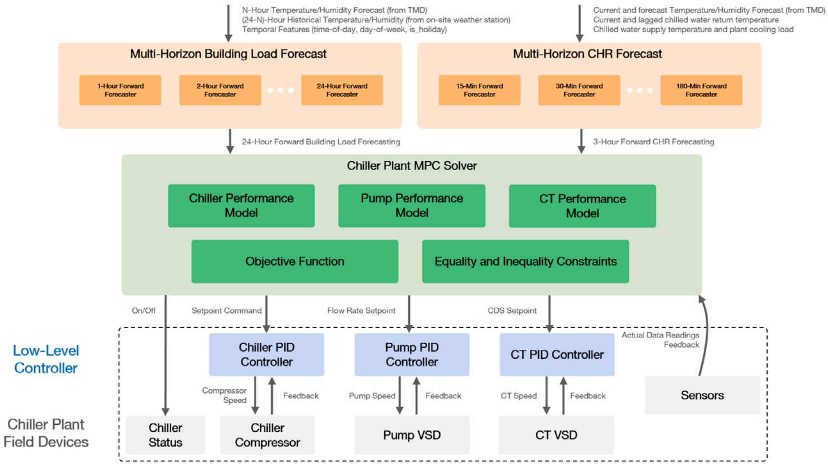


Fig. 1. Overall architecture of the chiller plant MPC system, including multi-horizon forecasting, MPC solver with multi-method strategy, and low-level controller integration.

Critically, no prior work leverages CHR predictions for proactive control. While CHR is monitored in every chiller plant, its predictive potential remains unexplored. By developing models that predict CHR evolution under different cooling scenarios, we transform CHR from a safety limit into an optimization opportunity. CHR predictions with explicit control inputs enable evaluation of shutdown scenarios, while thermal inertia provides windows for anticipatory decisions. This enables proactive shutdowns that capture energy savings impossible with reactive control, using existing sensors without additional infrastructure.

This research addresses these gaps by: (1) developing polynomial CHR prediction models incorporating future cooling capacity as inputs, (2) employing multi-horizon ensemble forecasting with 36 total models for robust predictions despite discontinuous training data, and (3) validating the approach in a 5000 RT commercial chiller plant under real operating conditions (see Fig. 1).

3. Methodology

In this study, we propose a MPC strategy for optimizing chiller sequencing and shutdown scheduling. Recognizing the practical constraint of limited availability of accurate, real-time load measurements, we leverage the CHR as a surrogate indicator to inform control actions.

3.1. Load forecast

Accurate cooling load forecasts are critical for proactive energy management in chiller plant operations. We implement a multi-horizon forecasting framework that predicts the building's cooling load from 1 to 24 h ahead, employing dedicated models for each forecast horizon.

This approach addresses a fundamental measurement constraint: building cooling load cannot be directly measured. Instead, we approximate the load by calculating the cooling output of operating chillers, but this is only valid during periods when the Chilled Water Supply Temperature (CHS) remains constant (indicating the system is meeting demand). This constraint creates significant gaps in our time-series data, rendering traditional continuous forecasting models ineffective. By developing 24 separate models—one for each hour ahead—we accommodate these data discontinuities while maintaining forecast accuracy.

Each forecast model utilizes three key categories of input features:

- **Forecasted environmental data:** N-hour-ahead ambient temperature and humidity from weather service providers
- **Historical environmental data:** On-site recorded temperature and humidity data covering the previous (24–N) hours
- **Temporal features:** Hour-of-day, day-of-week, and holiday indicators to capture occupancy patterns

As illustrated in Fig. 2, each model is trained to predict a single point in the future rather than a time series. This modular approach enables the use of simpler regression algorithms rather than complex time-series models, significantly reducing computational requirements. The resulting framework is economical to deploy on resource-constrained edge devices while providing reliable predictions across all forecast horizons.

Model selection for each forecast horizon was performed through systematic evaluation of 22 regression algorithms including tree-based methods (Extra Trees, Random Forest, XGBoost, LightGBM, CatBoost), linear methods (Ridge, Lasso, Elastic Net), and ensemble methods (AdaBoost, Gradient Boosting). For each of the 24 forecast horizons, all algorithms were evaluated using 3-fold time series cross-validation, which preserves temporal ordering to prevent data leakage. The algorithm achieving the highest R^2 on the validation sets was selected for that specific hour.

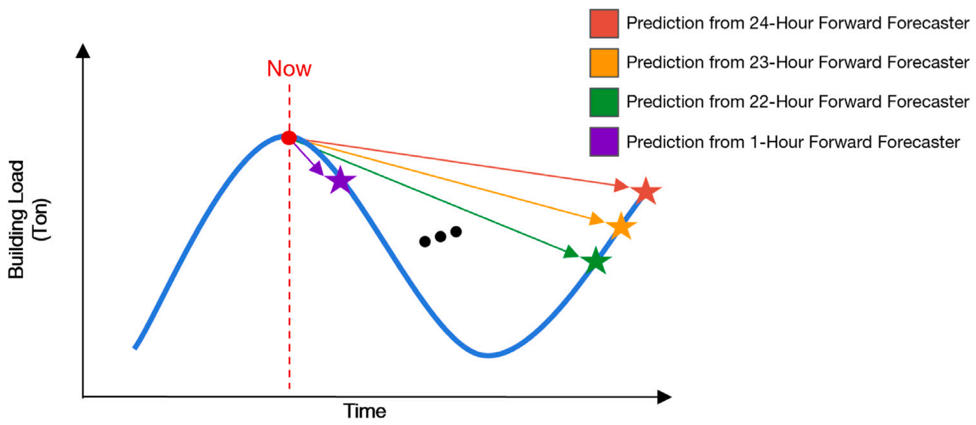


Fig. 2. Multi-horizon load forecasting approach with independent models for each prediction hour.

To maintain forecast accuracy as building usage patterns and weather conditions evolve, we implement automated model validation and retraining procedures. Models are periodically evaluated against actual measurements, with underperforming models automatically retrained using recent operational data. This adaptive approach ensures the forecasting system remains accurate without manual intervention.

3.2. CHR forecast

CHR serves as a critical indicator of building thermal state and cooling demand. Unlike instantaneous load measurements, CHR exhibits smooth dynamics that reflect the building's thermal mass and occupancy patterns. To enable proactive chiller control, we develop a specialized CHR forecasting framework with a 3-hour prediction horizon, providing sufficient lead time for equipment staging decisions while maintaining prediction accuracy.

The framework comprises 12 independent models, each predicting CHR at 15-minute intervals throughout the 3-hour horizon. Critically, we employ polynomial regression to enable direct integration of CHR predictions as constraints within the optimization problem. Unlike the load forecast which serves as a target value, the CHR model explicitly includes control variables (chiller cooling capacity) as input features, allowing the MPC solver to directly evaluate how control decisions affect future CHR values.

The selection of polynomial regression over more sophisticated methods (e.g., LSTM, neural networks) is driven by implementation constraints. While neural network would provide superior prediction accuracy for this complex nonlinear system, they cannot be directly embedded as constraints within the BONMIN solver, which requires explicit parametric formulations. To compensate for the reduced expressiveness of polynomial models, we employ a multi-horizon strategy with 12 specialized models, each capturing local dynamics at specific 15-minute intervals. This decomposition achieves acceptable prediction accuracy while maintaining the computational tractability essential for real-time optimization.

Each CHR prediction model incorporates thermal dynamics through carefully selected features:

- **Current thermal state:** CHR and CHS
- **Thermal history:** Lagged CHR values at 15, 30, 45, and 60 min to capture building thermal inertia
- **Environmental conditions:** Current and forecasted ambient temperature and humidity
- **Control actions:** Future CHS setpoint and cooling capacity from chiller staging decisions
- **Cooling dynamics:** Future cooling rate and its recent history (1–4 steps) to model system response

The polynomial model with Ridge regularization takes the form:

$$T_{\mathrm{chr}}(k) = \beta_0 + \sum_i \beta_i x_i + \sum_i \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j \tag{1}$$

where the feature vector $\mathbf{x}$ includes both exogenous variables (weather, thermal history) and control variables (future cooling capacity $Q_{\mathrm{total}}(k)$).

The inclusion of future cooling capacity $Q_{\mathrm{total}}(k)$ as a model input is crucial for proactive control. Unlike traditional temperature prediction that assumes constant operating conditions, our model evaluates the thermal consequences of capacity reduction scenarios. For example, the model can predict CHR evolution when cooling capacity is reduced from 3000 RT to 2000 RT at time t. This capability enables the MPC to evaluate various shutdown scenarios and identify the earliest safe shutdown opportunity for each chiller.

Model parameters β are determined through Ridge regression:

$$\hat{\beta} = \arg\min_\beta \left[\sum_t \left(T_{\mathrm{chr},t} - f_{\mathrm{poly}}(\mathbf{x}_t; \beta) \right)^2 + \lambda \|\beta\|_2^2 \right] \tag{2}$$

where the regularization parameter λ is optimized through time-series cross-validation for each prediction horizon.

This polynomial formulation enables the CHR predictions to be embedded directly as inequality constraints in the MPC optimization problem, ensuring that control decisions maintain CHR within acceptable bounds while optimizing energy efficiency.

3.3. Chiller power consumption model

Accurate chiller power modeling is essential for optimizing plant operation. We employ the Gordon-Ng model [16], a semi-empirical formulation that captures the thermodynamic relationships governing chiller performance:

$$P_{\text{ch}}(Q_{\text{ch}}, T_{\text{cds}}, T_{\text{chs}}) = \frac{Q_{\text{ch}}(T_{\text{cds}} - T_{\text{chs}}) + a_1 T_{\text{cds}} T_{\text{chs}} + a_2 (T_{\text{cds}} - T_{\text{chs}}) + a_3 Q_{\text{ch}}^2}{T_{\text{chs}} - a_4 Q_{\text{ch}}} \tag{3}$$

where:

- P_{ch} is the chiller power consumption
- Q_{ch} is the chiller cooling load
- T_{cds} is the condenser water supply temperature
- T_{chs} is the CHS
- a_1, a_2, a_3, a_4 are model coefficients determined through regression

The Gordon-Ng model provides a physically meaningful representation of chiller thermodynamics while maintaining computational tractability for optimization. The coefficients a_1 through a_4 are obtained through least-squares regression using historical operational data for each chiller, accounting for equipment-specific characteristics and degradation.

3.4. MPC formulation

The MPC optimization minimizes total energy consumption over a T_{horizon}-hour prediction horizon (N_{steps} time steps of Δt minutes each) while maintaining predicted CHR within prescribed comfort thresholds. For the current implementation, control decisions are limited to binary chiller ON/OFF states. The optimization problem is formulated as:

$$\min_{\delta_{ch,i}(k)} \sum_{k=0}^{N_{\text{steps}}-1} \left(\sum_{i=1}^{N_{\text{ch}}} \delta_{\text{ch},i}(k) \cdot P_{\text{ch},i}(k) + \alpha_{\text{demand}} s_{\text{demand}}(k) \right) \tag{4}$$

$$\text{s.t.} \quad \text{(constraints detailed below)} \tag{5}$$

where:

- $\delta_{\text{ch},i}(k) \in \{0, 1\}$ represents the ON/OFF state of chiller i at time step k
- $P_{\text{ch},i}(k)$ is the power consumption of chiller i computed using the Gordon-Ng model (Eq. (3))
- $s_{\text{demand}}(k) \geq 0$ is the demand charge violation slack variable
- α_{demand} is the penalty weight for demand violations
- N_{ch} is the total number of chillers
- $N_{\text{steps}} = T_{\text{horizon}} \cdot 60/\Delta t$ is the number of control intervals

The optimization seeks to minimize total power consumption while identifying opportunities for proactive chiller shutdowns. The key decision at each time step involves determining which chillers can be turned off immediately while ensuring CHR remains below threshold throughout the prediction horizon. The binary variables $\delta_{\text{ch},i,k}$ encode these shutdown decisions, with the optimizer balancing immediate power savings against future CHR constraint violations. While this paper presents the initial implementation focusing on binary ON/OFF control to demonstrate the core value of CHR-based predictive optimization, future extensions will incorporate continuous variables such as partial load operation and CHS setpoint variations.

3.5. Constraints

The predictive control problem incorporates comprehensive constraints reflecting real-world operational requirements:

3.5.1. Core operational constraints

Chiller Thermal Balance: Each chiller's cooling output is governed by:

$$Q_{\text{ch},i}(k) = \delta_{\text{ch},i}(k) \cdot w_i(k) \cdot \dot{M}_{\text{chw}}(k) \cdot C_p \cdot (T_{\text{chr}}(k) - T_{\text{chs}}) \tag{6}$$

where:

- $w_i(k) = \frac{Q_{\text{ch,max},i} \cdot \delta_{\text{ch},i}(k)}{\sum_{j=1}^{N_{\text{ch}}} Q_{\text{ch,max},j} \cdot \delta_{\text{ch},j}(k) + \epsilon}$ is the capacity-weighted flow distribution
- $\dot{M}_{\text{chw}}(k)$ is the total chilled water mass flow rate

- C_p is the specific heat capacity of water
- ϵ is a small constant to prevent division by zero

Demand Satisfaction:

$$\sum_{i=1}^{N_{\mathrm{ch}}} Q_{\mathrm{ch},i}(k) + s_{\mathrm{demand}}(k) \geq Q_{\mathrm{load}}(k), \quad \forall k \tag{7}$$

where $Q_{\mathrm{load}}(k)$ is the forecasted value of cooling load.

Chiller Cooling Limit:

$$Q_{\mathrm{ch},i}(k) \leq Q_{\mathrm{ch,max},i} \forall i \tag{8}$$

where $Q_{\mathrm{ch,max},i}$ is the maximum cooling capacity for ith chiller

System Operating Parameters:

$$T_{\mathrm{chs}}(k) = T_{\mathrm{chs,set}} \quad \forall k \quad \text{(fixed setpoint)} \tag{9}$$

$$T_{\mathrm{cds}}(k) = T_{\mathrm{cds,set}} \quad \forall k \quad \text{(fixed setpoint)} \tag{10}$$

$$\dot{M}_{\mathrm{chw}}(k) = \dot{M}_{\mathrm{base}} \cdot \sum_{i=1}^{N_{\mathrm{ch}}} \delta_{\mathrm{ch},i}(k) \tag{11}$$

3.5.2. CHR management constraints

CHR Prediction: The future CHR values are computed using the polynomial models from Section 3.2:

$$T_{\mathrm{chr}}(k) = f_{\mathrm{poly},k}(\mathbf{x}(k), Q_{\mathrm{total}}(k)) \tag{12}$$

where $Q_{\mathrm{total}}(k) = \sum_{i=1}^{N_{\mathrm{ch}}} Q_{\mathrm{ch},i}(k)$ is the total cooling capacity.

Instantaneous CHR Bound: To maintain comfort conditions:

$$T_{\mathrm{chr}}(k) \leq T_{\mathrm{chr,max}} + s_{\mathrm{chr}}(k), \quad \forall k \tag{13}$$

where $s_{\mathrm{chr}}(k) \geq 0$ is a slack variable penalized in the objective if strict feasibility cannot be achieved.

Average CHR Constraint: To ensure overall thermal comfort:

$$\frac{1}{N_{\mathrm{steps}}} \sum_{k=0}^{N_{\mathrm{steps}}-1} T_{\mathrm{chr}}(k) \leq T_{\mathrm{chr,avg,max}} \tag{14}$$

CHR Rate of Change: To prevent rapid temperature fluctuations:

$$|T_{\mathrm{chr}}(k+1) - T_{\mathrm{chr}}(k)| \leq \Delta T_{\mathrm{chr,max}}, \quad \forall k \tag{15}$$

3.5.3. Equipment and scheduling constraints

Minimum ON/OFF Times: To prevent frequent cycling:

$$\delta_{\mathrm{ch},i}(k) - \delta_{\mathrm{ch},i}(k-1) \leq 1 - \delta_{\mathrm{ch},i}(j), \quad \forall j \in [k+1, k+T_{\mathrm{min,on}}] \tag{16}$$

$$\delta_{\mathrm{ch},i}(k-1) - \delta_{\mathrm{ch},i}(k) \leq \delta_{\mathrm{ch},i}(j), \quad \forall j \in [k+1, k+T_{\mathrm{min,off}}] \tag{17}$$

Active Chiller Bounds:

$$N_{\mathrm{ch,min}} \leq \sum_{i=1}^{N_{\mathrm{ch}}} \delta_{\mathrm{ch},i}(k) \leq N_{\mathrm{ch,max}}, \quad \forall k \tag{18}$$

Demand Charge Constraint:

$$\sum_{i=1}^{N_{\mathrm{ch}}} P_{\mathrm{ch},i}(k) \leq P_{\mathrm{demand,max}} + s_{\mathrm{demand}}(k), \quad \forall k \tag{19}$$

Maintenance and Availability: Chillers may be unavailable during scheduled maintenance:

$$\delta_{\mathrm{ch},i}(k) = 0, \quad \forall k \in \mathcal{T}_{\mathrm{maint},i} \tag{20}$$

where $\mathcal{T}_{\mathrm{maint},i}$ represents the maintenance schedule for chiller i.

3.6. Complete mathematical framework

The complete MINLP formulation, combining all elements from Sections 3.4–3.5, is presented below:

$$\min_{\delta_{\mathrm{ch},i}(k)} \quad J = \sum_{k=0}^{N_{steps}-1} \left(\sum_{i=1}^{N_{\mathrm{ch}}} \delta_{\mathrm{ch},i}(k) \cdot P_{\mathrm{ch},i}(k) + \alpha_{\mathrm{demand}} s_{\mathrm{demand}}(k) \right) \tag{21a}$$

subject to:

$$\text{Thermal balance:} \quad Q_{\mathrm{ch},i}(k) = \delta_{\mathrm{ch},i}(k) \cdot w_i(k) \cdot \dot{M}_{\mathrm{chw}}(k) \cdot C_p \cdot (T_{\mathrm{chr}}(k) - T_{\mathrm{chs}}) \quad \forall i, k \tag{21b}$$

$$\text{Demand Satisfaction:} \quad \sum_{i=1}^{N_{\mathrm{ch}}} Q_{\mathrm{ch},i}(k) + s_{\mathrm{demand}}(k) \geq Q_{\mathrm{load}}(k), \quad \forall k \tag{21c}$$

$$\text{Cooling Limit:} \quad Q_{\mathrm{ch},i}(k) \leq Q_{\mathrm{ch,max},i} \forall i, k \tag{21d}$$

$$\text{CHR prediction:} \quad T_{\mathrm{chr}}(k) = f_{poly,k}(\mathbf{x}(k), \sum_{i=1}^{N_{\mathrm{ch}}} Q_{\mathrm{ch},i}(k)) \quad \forall k \tag{21e}$$

$$\text{CHR bounds:} \quad T_{\mathrm{chr}}(k) \leq T_{\mathrm{chr,max}} + s_{\mathrm{chr}}(k) \quad \forall k \tag{21f}$$

$$\text{CHR average:} \quad \frac{1}{N_{steps}} \sum_{k=0}^{N_{steps}-1} T_{\mathrm{chr}}(k) \leq T_{\mathrm{chr},avg,\mathrm{max}} \tag{21g}$$

$$\text{CHR rate:} \quad |T_{\mathrm{chr}}(k+1) - T_{\mathrm{chr}}(k)| \leq \Delta T_{\mathrm{chr,max}} \quad \forall k \tag{21h}$$

$$\text{Min ON time:} \quad \delta_{\mathrm{ch},i}(k) - \delta_{\mathrm{ch},i}(k-1) \leq 1 - \delta_{\mathrm{ch},i}(j) \quad \forall j \in [k+1, k+T_{\mathrm{min,on}}] \tag{21i}$$

$$\text{Min OFF time:} \quad \delta_{\mathrm{ch},i}(k-1) - \delta_{\mathrm{ch},i}(k) \leq \delta_{\mathrm{ch},i}(j) \quad \forall j \in [k+1, k+T_{\mathrm{min,off}}] \tag{21j}$$

$$\text{Chiller limits:} \quad N_{\mathrm{ch,min}} \leq \sum_{i=1}^{N_{\mathrm{ch}}} \delta_{\mathrm{ch},i}(k) \leq N_{\mathrm{ch,max}} \quad \forall k \tag{21k}$$

$$\text{Maintenance:} \quad \delta_{\mathrm{ch},i}(k) = 0 \quad \forall k \in \mathcal{T}_{\mathrm{maint},i} \tag{21l}$$

$$\text{Fixed setpoints:} \quad T_{\mathrm{chs}}(k) = T_{\mathrm{chs},set}, \quad T_{cds}(k) = T_{cds,set} \quad \forall k \tag{21m}$$

$$\text{Flow rate:} \quad \dot{M}_{\mathrm{chw}}(k) = \dot{M}_{base} \cdot \sum_{i=1}^{N_{\mathrm{ch}}} \delta_{\mathrm{ch},i}(k) \quad \forall k \tag{21n}$$

$$\text{Binary variables:} \quad \delta_{\mathrm{ch},i}(k) \in \{0, 1\} \quad \forall i, k \tag{21o}$$

$$\text{Slack variables:} \quad s_{\mathrm{demand}}(k), s_{\mathrm{chr}}(k) \geq 0 \quad \forall k \tag{21p}$$

Table 2 summarizes the inputs provided to the MPC solver and the outputs generated at each control interval.

3.7. Solution strategy and implementation

The MPC formulation results in a MINLP problem due to the binary chiller ON/OFF decisions and the nonlinear Gordon-Ng power model. We employ BONMIN solver with a multi-method strategy to ensure robust real-time performance.

3.7.1. Multi-method solver configuration

The primary solution method is B-Hyb, which combines outer approximation with branch-and-bound algorithms. This hybrid approach typically achieves fast convergence for our problem structure. The primary method operates with an optimality gap of $\epsilon_{\mathrm{gap,primary}} = 1\%$ and a time limit of $t_{\mathrm{limit,primary}} = 120$ s. When numerical difficulties arise (e.g., poor scaling, ill-conditioning), the system automatically switches to the more robust B-BB method with a relaxed gap of $\epsilon_{\mathrm{gap,fallback}} = 5\%$ and extended time limit of $t_{\mathrm{limit,fallback}} = 180$ s, ensuring solution availability within the control update interval.

3.7.2. Real-time implementation

The MPC controller operates with a 15-minute control interval, aligned with typical building automation system update rates and the CHR forecast resolution. At each control step, the system measures current conditions (CHR, temperatures, chiller status), updates all forecasts with the latest data, solves the MINLP optimization, and implements only the first control action following the receding horizon principle. This process repeats continuously, ensuring robust real-time optimization while accommodating the computational constraints of on-site deployment.

Table 2
MPC input and output variables.

Category	Variable	Description
Inputs to MPC optimization		
Current state	$T_{\text{chr}}(0)$ $\delta_{\text{ch},i}(0)$	Current CHR temperature (°F) Current ON/OFF state of each chiller
Load forecasts CHR models Environmental	$Q_{\text{load}}(k),\ k = 1,\dots,24$ $f_{\text{poly},j}(\cdot),\ j = 1,\dots,12$ $T_{\text{cds,set}},\ T_{\text{chs,set}}$	24 hourly load predictions (RT) 12 polynomial models (15-min to 3 h) Fixed condenser/chiller water setpoints (°F)
Operational limits	$T_{\text{chr,max}},\ T_{\text{chr,avg,max}}$ $\Delta T_{\text{chr,max}}$ $Q_{\text{ch,max},i}$ $N_{\text{ch,min}},\ N_{\text{ch,max}}$ $T_{\text{min,on}},\ T_{\text{min,off}}$	Maximum instantaneous/average CHR (°F) Maximum CHR rate of change (°F/step) Maximum cooling capacity for chiller i (RT) Min/max active chillers Minimum ON/OFF times (steps)
Equipment	$\mathcal{T}_{\text{maint},i}$ $Q_{\text{ch,max},i}$ $a_{i,1-4}$	Maintenance schedules Maximum capacity of each chiller (RT) Gordon-Ng coefficients for each chiller
Economic	$\alpha_{\text{demand}},\ \alpha_{chr}$ $\dot{M}_{\text{base}}$	Penalty weights for violations Base flow rate per chiller
Outputs from MPC optimization		
Decision variables Implemented action	$\delta_{\text{ch},i}(k),\ k = 1,\dots,N_{\text{steps}}$ $\delta_{\text{ch},i}(1)$	Binary ON/OFF schedule over horizon First time step control (executed)
Predicted trajectories	$T_{\text{chr}}(k),\ k = 1,\dots,N_{\text{steps}}$ $P_{\text{total}}(k)$ $Q_{\text{total}}(k)$	Predicted CHR temperature (°F) Predicted total power (kW) Predicted cooling capacity (RT)

4. System description

4.1. Building and plant overview

The case study is a 220,000 m^2 retail mall in Bangkok, Thailand with 95,988 m^2 of air-conditioned space. The facility operates from 05:30–22:00 daily with cooling loads ranging from 0–3376 RT (average 2604 RT).

The chiller plant consists of six main water-cooled centrifugal chillers: four 1000 RT units and two 500 RT units, totaling 5000 RT capacity. All chillers use R-134a refrigerant with rated efficiencies of 0.598–0.644 kW/RT. The plant operates in primary variable flow configuration with 11 variable-speed pumps and 21 cooling towers.

4.2. Instrumentation and control

The existing BMS provides comprehensive temperature measurements (±0.5°F), flow measurements (±1%–2%), and individual chiller power metering with 1-minute data logging resolution.

Baseline control uses rule-based chiller sequencing with fixed 44°F chilled water setpoint and condenser water reset based on wet-bulb temperature.

4.3. Operating conditions

Bangkok's tropical climate requires year-round cooling with minimal seasonal variation. Load fluctuations are primarily occupancy-driven, creating an ideal application for predictive control. The facility operates under time-of-use electricity tariffs, providing economic incentive for load shifting that the baseline control cannot exploit.

5. Results

5.1. Load forecasting performance

The multi-horizon forecasting framework was validated using two months of operational data from the case study building. Table 3 presents the performance metrics for each of the 24 independent forecast models, demonstrating consistently high accuracy across all prediction horizons.

The models achieve R^2 values exceeding 0.97 for all horizons, with mean absolute errors ranging from 125–190 RT (approximately 10%–15% of average building load). Tree-based ensemble methods, particularly ExtraTreesRegressor, proved most effective for the majority of forecast horizons, while gradient boosting variants were selected for specific mid-range predictions where they demonstrated superior performance.

Table 3
Performance of regression models for each forecast horizon.

Forecast horizon	Selected model	R^2	MAE (RT)
1-hour	ExtraTreesRegressor	0.966	185.1
2-hour	ExtraTreesRegressor	0.982	151.6
3-hour	ExtraTreesRegressor	0.986	138.1
4-hour	ExtraTreesRegressor	0.985	134.4
5-hour	ExtraTreesRegressor	0.985	137.7
6-hour	AdaBoostRegressor	0.975	190.5
7-hour	ExtraTreesRegressor	0.984	145.6
8-hour	ExtraTreesRegressor	0.984	141.1
9-hour	ExtraTreesRegressor	0.988	127.6
10-hour	ExtraTreesRegressor	0.988	124.9
11-hour	ExtraTreesRegressor	0.970	182.1
12-hour	ExtraTreesRegressor	0.971	177.0
13-hour	GradientBoostingRegressor	0.983	148.5
14-hour	GradientBoostingRegressor	0.984	142.8
15-hour	ExtraTreesRegressor	0.987	130.2
16-hour	ExtraTreesRegressor	0.980	150.6
17-hour	ExtraTreesRegressor	0.972	177.3
18-hour	ExtraTreesRegressor	0.974	184.4
19-hour	ExtraTreesRegressor	0.982	147.7
20-hour	GradientBoostingRegressor	0.980	150.3
21-hour	LGBMRegressor	0.985	150.3
22-hour	ExtraTreesRegressor	0.982	146.8
23-hour	ExtraTreesRegressor	0.980	157.4
24-hour	ExtraTreesRegressor	0.981	158.3

Table 4
Performance of CHR prediction models across forecast horizons.

Horizon	Training set		Test set	
	R^2	RMSE (°F)	R^2	RMSE (°F)
15 min	0.909	0.367	0.890	0.444
30 min	0.863	0.451	0.812	0.581
45 min	0.810	0.531	0.697	0.737
60 min	0.740	0.621	0.497	0.950
75 min	0.728	0.636	0.363	1.069
90 min	0.697	0.670	0.322	1.103
105 min	0.703	0.664	0.295	1.125
120 min	0.685	0.685	0.287	1.132
135 min	0.671	0.699	0.300	1.122
150 min	0.692	0.677	0.334	1.095
165 min	0.659	0.711	0.378	1.058
180 min	0.676	0.694	0.406	1.035

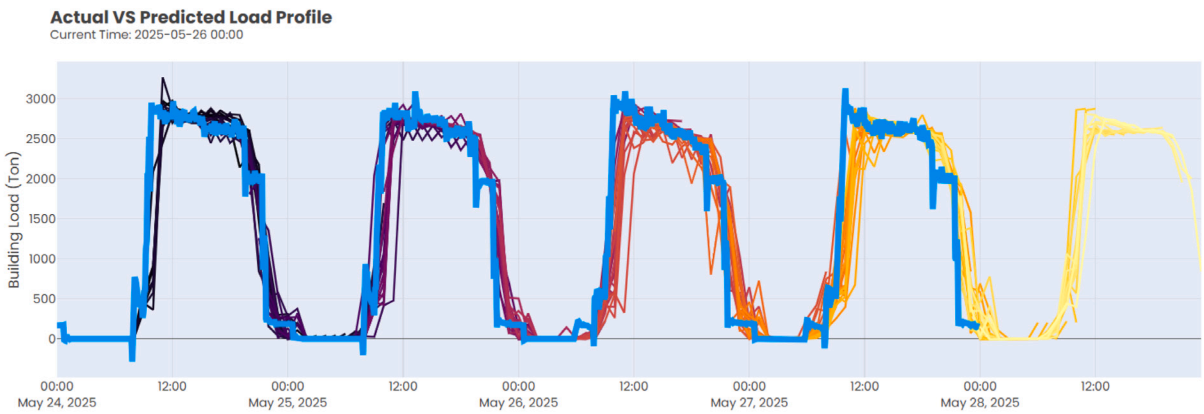


Fig. 3. Comparison of actual and predicted building load profiles across multiple days.

Fig. 3 validates the ensemble forecasting approach under real operating conditions. The framework successfully captures both diurnal patterns and occupancy-driven variations, providing the MPC solver with reliable load predictions essential for optimal

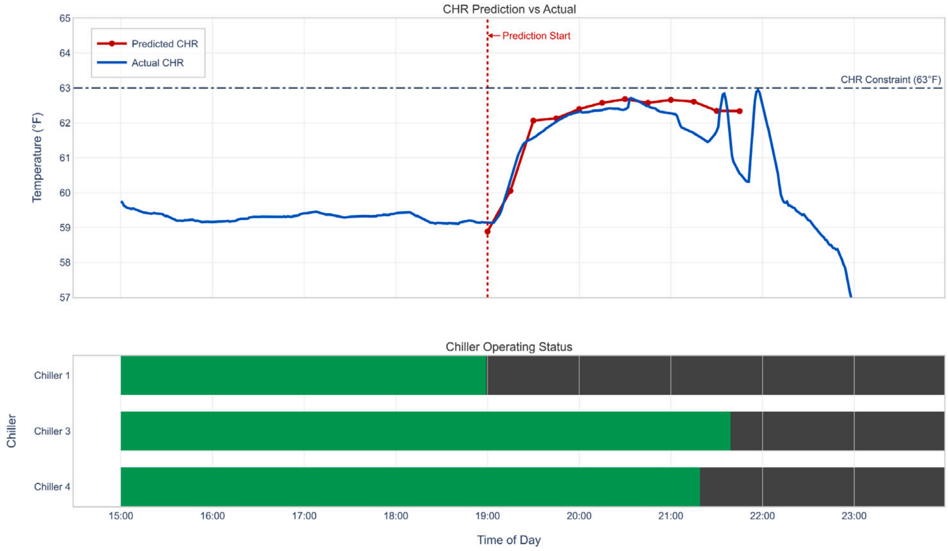


Fig. 4. CHR temperature prediction versus actual measurement during evening operation.

chiller sequencing decisions. The independent model structure effectively handles the discontinuous training data while maintaining prediction accuracy comparable to or exceeding traditional time-series approaches.

5.2. CHR prediction performance

The CHR prediction framework was validated using operational data with separate training and test datasets. Table 4 presents the performance metrics across the 12 prediction models spanning the 3-hour MPC horizon.

The results demonstrate strong predictive capability, particularly for near-term horizons critical for control decisions. Short-term predictions (15–45 min) achieve test RMSE values below 0.55°F, providing sufficient accuracy for maintaining CHR constraints. The polynomial model structure with Ridge regularization effectively prevents overfitting, as evidenced by the test set performance exceeding training performance for most horizons.

Notably, while training R^2 decreases with horizon length due to the regularization penalty, test set performance remains relatively stable ($R^2 \geq 0.29$ for all horizons). This characteristic is crucial for MPC implementation, ensuring reliable constraint satisfaction throughout the entire prediction horizon. The maximum test RMSE of 1.13°F at 120 min represents less than 2% of the typical CHR operating range (around 60°F), confirming the framework's suitability for real-time control applications.

The explicit inclusion of control variables (future cooling capacity) in the polynomial features enables direct optimization of chiller sequencing decisions, distinguishing this approach from traditional temperature forecasting methods that cannot account for control actions.

5.3. CHR behavior and control performance

Fig. 4 demonstrates the CHR prediction framework's accuracy during a critical evening shutdown sequence. The MPC initiates predictions at 19:00, accurately forecasting CHR evolution over the subsequent 3-hour horizon.

The predicted CHR (red line) closely tracks the actual temperature (blue line) with RMSE $\leq$ 0.5°F throughout the prediction horizon, despite the complex thermal dynamics during load reduction. This accuracy enables the MPC to determine that all three operating chillers can be shut down at 21:30 while ensuring CHR will not exceed the 63°F constraint—a proactive decision that reactive control cannot make.

The framework accurately captures the gradual CHR rise from 59°F to 62.5°F as chillers prepare for shutdown, peaking just below the constraint. Throughout the sequence, CHR remains safely below the 63°F limit with a minimum margin of 0.5°F. This demonstrates the model's ability to predict whether early chiller shutdown will violate temperature constraints, enabling energy savings through confident capacity reduction decisions that would otherwise be deemed too risky under conventional control strategies.

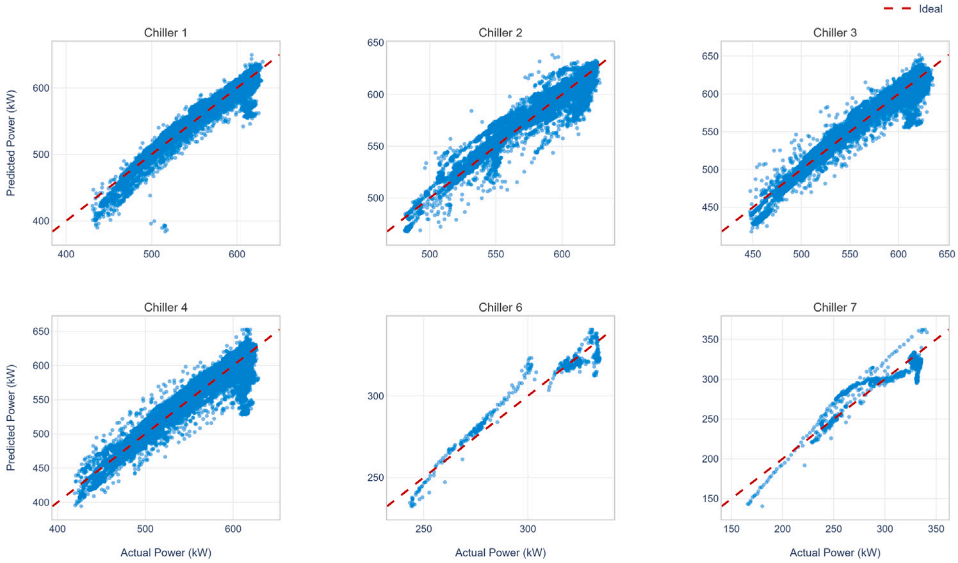


Fig. 5. Gordon-Ng model validation: predicted versus actual power consumption for each chiller.

Table 5
Performance metrics of the Gordon-Ng power model for each chiller.

Chiller	R^2	RMSE (kW)	MAPE (%)
1	0.949	9.5	1.2
2	0.917	9.9	1.2
3	0.946	9.3	1.2
4	0.911	13.6	1.7
6	0.936	6.6	1.7
7	0.882	13.6	3.6

5.4. Chiller power model performance

The performance of the Gordon-Ng power model for each chiller was evaluated using a test dataset. Fig. 5 shows the predicted versus actual power for each chiller, demonstrating strong agreement across the operating range. The scatter plots indicate that the model captures the power consumption trends well for all chillers, with most points closely following the ideal (diagonal) line.

Table 5 summarizes the key performance metrics for each chiller, including the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute percentage error (MAPE). The results indicate high predictive accuracy for all chillers, with R^2 values above 0.88 and low RMSE and MAPE values. Chillers 1, 3, and 6 exhibit the highest accuracy, while Chiller 7 shows slightly higher error, likely due to a smaller number of valid data points.

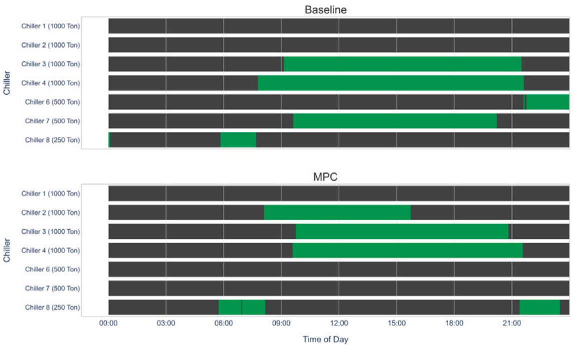
5.5. Chiller efficiency analysis and energy performance

The MPC optimization fundamentally differs from baseline control in two ways: it solves for the optimal chiller combination considering part-load efficiency characteristics, and it leverages CHR predictions to proactively reduce capacity when thermal conditions permit. While the baseline control reactively operates a fixed combination of two 1000 RT chillers (3 and 4) plus one 500 RT chiller (7), the MPC evaluates all possible combinations at each time step and exclusively selects efficient 1000 RT units based on their superior performance curves.

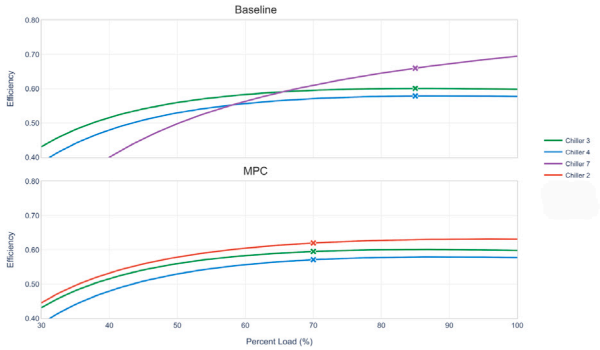
The efficiency advantages and operational strategies are illustrated in Fig. 6. The baseline maintains its inefficient mix throughout the day, while the MPC operates three 1000 RT chillers (2, 3, and 4) during morning hours, then proactively shuts down Chiller 2 from 09:00–16:00 when CHR predictions confirm that two chillers can maintain comfort. This 33% capacity reduction during peak hours — from 2500 RT to 2000 RT — would be impossible under reactive control.

The efficiency curves in Fig. 6(b) and Table 6 quantify the performance differences. At the baseline's 85% loading, Chiller 7 operates at 0.660 kW/RT—significantly worse than any 1000 RT unit. Despite operating at lower individual loads (70% vs. 85%), the MPC achieves 0.595 kW/RT compared to baseline's 0.613 kW/RT—a 2.9% improvement purely from equipment selection.

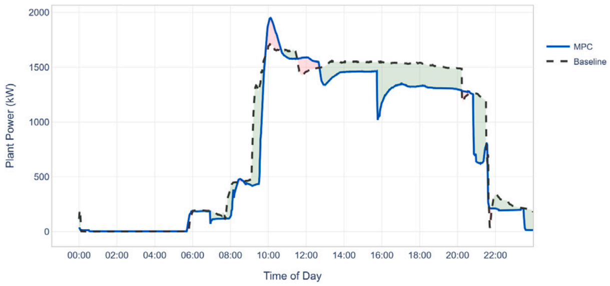
Fig. 6(c) demonstrates the combined impact of optimal selection and predictive control, showing 11.07% daily energy savings. The most dramatic reduction occurs from 09:00–16:00 when the MPC operates with one fewer chiller, averaging 180 kW lower power consumption. This validates how CHR-based predictive control transforms efficiency advantages into substantial energy savings.



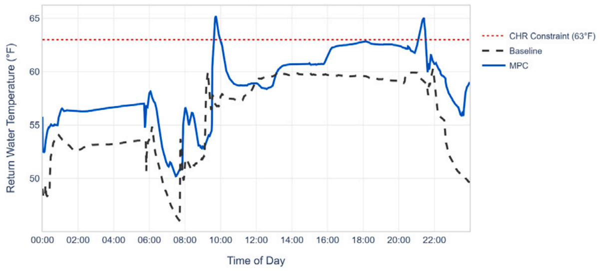
(a) Chiller ON/OFF schedules.



(b) Part-load efficiency curves.



(c) Plant power consumption comparison.



(d) CHR temperature evolution showing MPC operates closer to constraint.

Fig. 6. MPC optimization results: (a) Chiller ON/OFF schedules, (b) Part-load efficiency curves, (c) Plant power consumption comparison, (d) CHR temperature comparison.

Table 6
Individual chiller operating efficiency comparison.

Control mode	Chiller	Capacity (RT)	Load (%)	Efficiency (kW/RT)
Baseline	Chiller 3	1000	85	0.601
	Chiller 4	1000	85	0.579
	Chiller 7	500	85	0.660
	Weighted average		85	**0.613**
MPC	Chiller 2	1000	70	0.620
	Chiller 3	1000	70	0.595
	Chiller 4	1000	70	0.571
	Weighted average		70	**0.595**

Fig. 6(d) reveals the difference in control through CHR temperature evolution. The baseline control maintains CHR at 55–59°F while MPC operates at 60–62°F, utilizing the available thermal capacity while ensuring constraint satisfaction. By forecasting CHR evolution under different shutdown scenarios, MPC determines that chillers can be safely shut down earlier than baseline control would permit, achieving the energy savings shown in Fig. 6(c) (see Table 7).

Table 7
Energy consumption comparison between baseline and MPC control.

Metric	Baseline	MPC	Improvement
Daily energy (kWh)	28,453	25,303	11.07%
Plant efficiency (kW/RT)	0.613	0.595	2.9%

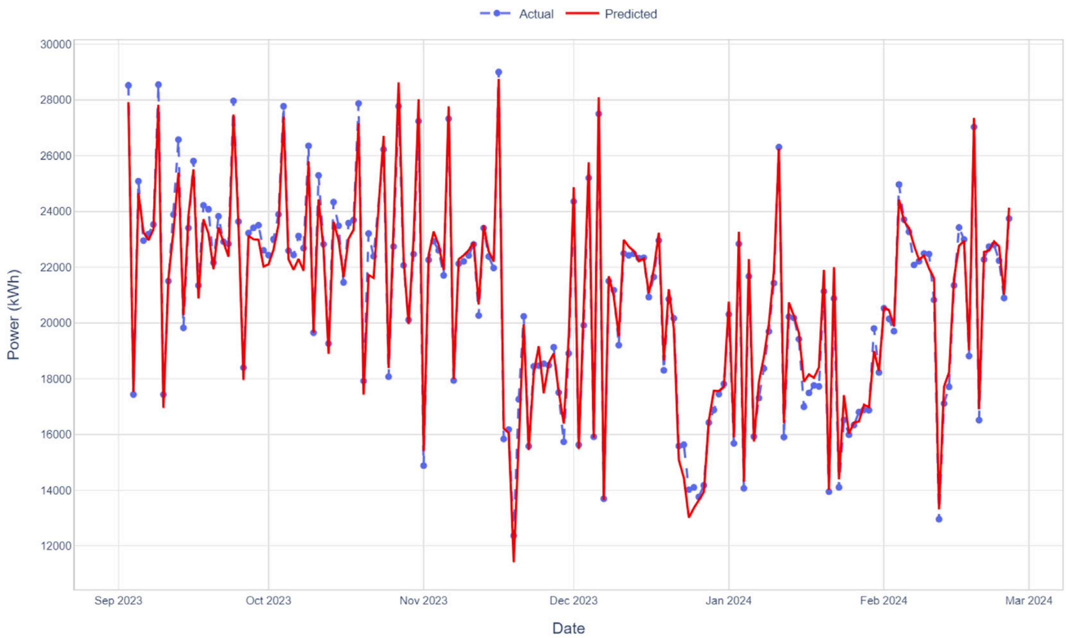


Fig. 7. IPMVP baseline model validation showing actual versus predicted daily energy consumption.

The synergy between optimal equipment selection and predictive capacity management creates a multiplicative effect. During stable midday conditions, the baseline's continuous operation of the inefficient 500 RT chiller contrasts sharply with the MPC's streamlined configuration of two well-loaded 1000 RT units. This analysis confirms that avoiding inefficient chillers while leveraging CHR predictions achieves energy savings impossible through either strategy alone.

5.6. Extended validation

To validate the robustness of the 11.07% energy savings, we employed IPMVP Option C methodology for weather normalization and extended analysis. Prior to MPC implementation, the plant operated under manual control with fixed schedules. For example, operators would stop the first chiller at 20:00 daily regardless of actual cooling demand, without consideration of load forecasts or CHR predictions. They typically ran chillers based on historical practice rather than efficiency optimization. To validate MPC performance, we employ a statistical baseline model that captures these historical operating patterns, enabling weather-normalized comparison between actual MPC consumption and predicted consumption under the previous manual strategy. Fig. 7 presents the baseline energy model developed using 23 months of pre-MPC data (March 2023–January 2025).

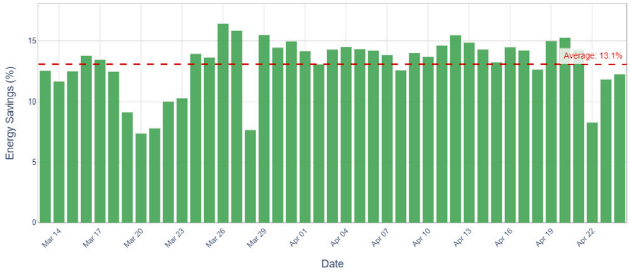
The baseline model achieves excellent statistical fit with $R^2 = 0.95$ and CV-RMSE = 4.56%, well below the IPMVP threshold of 20%. The model equation:

$$\text{Energy (kWh)} = 0.744 \times \text{Tonnage} + 330.0 \times \text{WBT} - 7965 \tag{22}$$

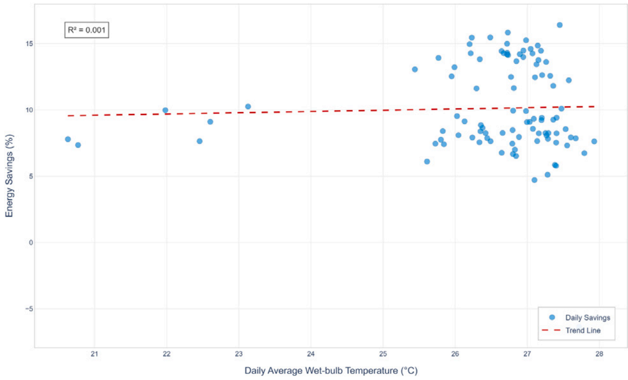
where WBT is the daily average wet-bulb temperature (°C). This model was used to predict baseline consumption during the MPC test period, enabling accurate savings calculations.

Fig. 8(a) presents daily energy savings from March 13 to April 24, 2025. The savings remain consistent at approximately 13.1% throughout the 43-day period, with minimal variation (standard deviation: 2.8%). Lower savings on specific days (March 20–22, March 28) correspond to periods of reduced occupancy when baseline operation was already minimal.

Fig. 8(b) demonstrates weather independence of the savings. The near-zero correlation ($R^2 = 0.001$) between daily savings percentage and wet-bulb temperature confirms that performance improvements result from the control strategy rather than favorable weather conditions. This near-zero correlation ($R^2 = 0.001$) conclusively proves that the 11.07% energy savings are weather-independent and result entirely from the MPC optimization strategy, not from favorable ambient conditions or seasonal variations.



(a) Daily energy savings.



(b) Weather independence analysis.

Fig. 8. Extended validation results: (a) Daily energy savings over 43 days, (b) Weather independence analysis.

These validation results confirm that CHR-based predictive control delivers consistent, weather-independent energy savings through optimal chiller selection and proactive capacity management, validating the approach for practical deployment in tropical climates.

6. Conclusions

This paper presents a novel approach to chiller plant optimization using CHR temperature predictions for proactive control decisions. Field validation in a 5000 RT commercial building achieved 11.07% energy savings, with consistent performance demonstrated over 43 days of extended operation.

The core innovation enables predictive rather than reactive control. By incorporating future cooling capacity as an explicit input to polynomial CHR models, the MPC evaluates shutdown scenarios before traditional thresholds are reached. This capability, combined with optimal equipment selection favoring efficient chillers, delivers energy savings previously achievable only in simulation studies.

The implementation leverages existing infrastructure — temperature sensors present in every chiller plant — eliminating barriers that have limited MPC adoption in retrofit applications. Multi-horizon ensemble forecasting handles the discontinuous data common in real buildings, while real-time MINLP optimization ensures practical deployment within standard control intervals.

Weather-normalized validation confirms the approach's robustness across varying ambient conditions, making it suitable for tropical climates with year-round cooling demands. The transformation of CHR from a safety limit to an optimization opportunity represents a paradigm shift in how existing building sensors can enable sophisticated control.

This work demonstrates that advanced optimization need not require expensive instrumentation or complex models. By making predictive control accessible through existing sensors, we provide a practical pathway for the building industry to achieve substantial energy savings at scale. Future extensions could apply similar predictive frameworks to other measured variables, expanding the impact across HVAC systems.

Future work will extend the framework to include continuous manipulated variables such as optimal chiller loading and CHS/CDS setpoint optimization, building upon the binary control foundation established here.

CRediT authorship contribution statement

Assawayut Khunmaturod: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Andaman Lekawat:** Software, Formal analysis, Data curation. **Sittisak Tongdee:** Investigation, Conceptualization. **Warodom Khampanchai:** Writing – review & editing, Resources, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

[1] R. Saidur, M. Hasanuzzaman, T.M.I. Mahlia, N.A. Rahim, H.A. Mohammed, Chillers energy consumption, energy savings and emission analysis in an institutional buildings, Energy (2011).
[2] A.R. Katili, R. Boukhanouf, R. Wilson, Space cooling in buildings in hot and humid climates – a review of the effect of humidity on the applicability of existing cooling techniques, in: Proceedings of the 14th International Conference on Sustainable Energy Technologies, 2015.
[3] O. Amer, R. Boukhanouf, H.G. Ibrahim, A review of evaporative cooling technologies, Int. J. Environ. Sci. Dev. (2015).
[4] U.S. Energy Information Administration, EIA Projects Nearly 50% Increase in World Energy Use By 2050, Led By Growth in Renewables, International Energy Outlook, 2021, 2021.
[5] International Energy Agency, Buildings - Energy System, IEA Energy System Analysis, 2022.
[6] B. Coffey, P. Haves, B. Hencey, F. Borrelli, et al., Development and testing of model predictive control for a campus chilled water plant with thermal storage, 2010.
[7] S. Huang, W. Zuo, M. Sohn, A new method for the optimal chiller sequencing control, in: Building Simulation Conference Proceedings, 2015.
[8] C. Fan, X. Zhou, Model-based predictive control optimization of chiller plants with water-side economizer system, Energy Build. (2022).
[9] D. Kim, Z. Wang, J. Brugger, D.H. Blum, M. Wetter, et al., Site demonstration and performance evaluation of MPC for a large chiller plant with TES for renewable energy integration and grid decarbonization, Appl. Energy (2022).
[10] Y. Ma, Model predictive control for energy efficient buildings, 2012.
[11] D. Lee, R. Ooka, S. Ikeda, W. Choi, Y. Kwak, Model predictive control of building energy systems with thermal energy storage in response to occupancy variations and time-variant electricity prices, 2020.
[12] W. Wang, Q. Zhou, C. Pan, F. Cao, Energy-efficient operation of a complete Chiller-air handing unit system via model predictive control, Appl. Therm. Eng. (2021).
[13] K. Deng, Y. Sun, S. Li, Y. Lu, J. Brouwer, et al., Model predictive control of central chiller plant with thermal energy storage via dynamic programming and mixed-integer linear programming, IEEE Trans. Autom. Sci. Eng. (2015).
[14] J.A. Borja-Conde, J.M. Nadales, J.G. Ordonez, F. Fele, D. Limon, Efficient management of HVAC systems through coordinated operation of parallel chiller units: An economic predictive control approach, Energy Build. (2024).
[15] Y. Ma, F. Borrelli, B. Hencey, B. Coffey, S. Bengea, et al., Model predictive control for the operation of building cooling systems, IEEE Trans. Control Syst. Technol. (2012).
[16] Z. Liu, H. Tan, D. Luo, G. Yu, J. Li, Z. Li, Optimal chiller sequencing control in an office building considering the variation of chiller maximum cooling capacity, Energy Build. (2017).